

Machine learning to identify clinically relevant *Candida* yeast species

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Abstract

Fungal infections, especially due to *Candida* species, are on the rise. Multi-drug resistant organisms such as *Candida auris* are difficult and time consuming to identify accurately. Machine learning is increasingly being used in health care, especially in medical imaging. In this study, we evaluated the effectiveness of six convolutional neural networks (CNNs) to identify four clinically important *Candida* species. Wet-mounted images were captured using bright field live-cell microscopy followed by separating single-cells, budding-cells, and cell-group images which were then subjected to different machine learning algorithms (custom CNN, VGG16, ResNet50, InceptionV3, EfficientNetB0, and EfficientNetB7) to learn and predict *Candida* species. Among the six algorithms tested, the InceptionV3 model performed best in predicting *Candida* species from microscopy images. All models performed poorly on raw images obtained directly from the microscope. The performance of all models increased when trained on single and budding cell images. The InceptionV3 model identified budding cells of *C. albicans*, *C. auris*, *C. glabrata* (*Nakaseomyces glabrata*), and *C. haemulonii* in 97.0%, 74.0%, 68.0%, and 66.0% cases, respectively. For single cells of *C. albicans*, *C. auris*, *C. glabrata*, and *C. haemulonii* InceptionV3 identified 97.0%, 73.0%, 69.0%, and 73.0% cases, respectively. The sensitivity and specificity of InceptionV3 were 77.1% and 92.4%, respectively. Overall, this study provides proof of the concept that microscopy images from wet-mounted slides can be used to identify *Candida* yeast species using machine learning quickly and accurately.

Lay summary

Fungal infections due to *Candida* yeasts are increasing worldwide. Existing methods to identify these pathogens are difficult and time consuming. We find that machine learning can identify *Candida* species from images quickly and accurately, improving the diagnosis of infectious fungal diseases.

Key words: *Candida* species, deep neural networks, fungal infections, machine learning, medical AI diagnosis.

Introduction

Fungi cause serious infections affecting more than one billion people of all ages across the globe and are responsible for approximately 1.6 million deaths per year.¹ *Candida* species is the fourth leading agent to cause hospital-acquired bloodstream infections in USA², seventh in Europe,³ and affects 250 000 people worldwide and causes 50 000 deaths every year.⁴ *Candida* species, commonly found as commensals in human skin and mucosal surfaces, cause opportunistic infections when the immune system is impaired.⁵ The severity of the infection depends on the immune status of the individual, the infecting agent, and their virulence factors along with their ability to evade host factors.⁶ The mortality rate due to candidemia ranges from 22% to 75%.⁷

Nearly 90% of invasive *Candida* infections are caused by *Candida albicans*, *C. glabrata* (which has an alternate taxonomic name of *Nakaseomyces glabrata*⁸), *C. tropicalis*, *C. parapsilosis*, and *C. krusei*.⁹ Due to the limited antifungal treatment options to treat invasive infections,¹⁰ early diagnosis and prompt antifungal therapy are critical for successful clinical outcomes. *Candida auris* has emerged as a multi-drug resistant pathogen causing hospital-acquired infections and outbreaks. First reported in Japan in 2009, *C. auris* spread across the globe and recently the World Health

Organization declared this pathogen as a ‘critical priority’ pathogen.^{11–13} *Candida auris* is commonly encountered in prolonged hospitalized patients and is known to cause superficial to invasive bloodstream infections.¹⁴ The mortality rate due to this pathogenic yeast ranges from 30% to 60%.¹⁵ In some hospitals in India, *C. auris* is the most isolated *Candida* species from blood culture.¹⁶ Previous studies indicate that *C. auris* exhibits increased minimum inhibitory concentrations (MICs) to all three major classes of antifungal drugs to treat invasive infections (i.e., azoles, polyenes, and echinocandins).¹⁷ Several studies have reported that *C. auris* is often misidentified as *C. haemulonii*, *C. famata*, *C. sake*, and *Saccharomyces cerevisiae* by conventional laboratory methods (e.g., germ tube and carbohydrate fermentation tests) as well as automated commercial methods (e.g., VITEK and matrix-assisted laser desorption ionization-time of flight mass spectroscopy [MALDI-TOF MS]).^{18–20} Diagnostic-based antifungal treatments can reduce costs and shorten hospital stays compared to empirical treatments.²¹ Therefore, early detection of pathogenic *Candida* species can help improve patient outcomes and contain infectious fungal diseases.

Another major problem in the diagnosis of candidiasis is the delay in turnaround time from conventional methods. A few studies have utilized molecular techniques to

Received: September 19, 2023. Revised: December 6, 2023. Accepted: December 19, 2023

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reduce the turnaround time for the diagnosis of *Candida* infections. For example, Maaroufi et al.²² developed a real-time polymerase chain reaction (PCR)-based assay for the early detection and identification of commonly encountered *Candida* species from simulated blood cultures. This method showed high efficiency as an adjunct to blood culture systems with a short turnaround time (1–2 h) compared to traditional methods. Farina et al.²³ evaluated the performance of MALDI-TOF MS in identifying clinically relevant *Candida* species directly from positive blood cultures. They found that MALDI-TOF MS had a prediction accuracy of 90% compared to conventional methods, with a much shorter turnaround time (12–24 h). In addition to molecular methods, other techniques such as peptide nucleic acid fluorescence in situ hybridization (PNA-FISH),²⁴ flow cytometry,²⁵ and Raman spectroscopy²⁶ have been explored in the context of rapidly identifying *Candida* species. However, these methods require sophisticated instruments (e.g., MALDI-TOF MS, DNA sequencer, etc.), extensive laboratory expertise, or time-consuming preprocessing of *Candida* species.

The ‘gold standard’ for the diagnosis of the invasive candidiasis has been elaborated by Mycoses Study Group Education and Research Consortium group.²⁷ In their consensus guidelines, four tests are stated to confirm the diagnosis of invasive candidiasis: histopathology findings, positive culture from sterile site samples, detection of yeast from paraffin-embedded tissue samples, and blood culture positivity (most employed test). Other non-culture methods employed for the diagnosis of the invasive candidiasis are beta-D-glucan,²⁸ *C. albicans* germ tube antibody,²⁹ nucleic acid amplification test from blood samples,³⁰ and T2 Biosystems (*Candida* panel)-based diagnostic tests.³¹ However, these non-culture methods require expensive equipment and extensive expertise, limiting their usage in underdeveloped countries and in rural settings.³² Identifying these infecting *Candida* species is crucial due to the high diversity among the genus and the possibility of intrinsic and acquired resistance in some of these species.³³ The major drawback of these tests is their long turnaround times, which is approximately 72–96 h, leading to delays in antifungal treatment resulting in increased mortality.³⁴ Antifungal susceptibility testing takes an additional 48–72 h to provide the susceptibility profile of the infectious agent, further delaying antifungal therapy.

Artificial Intelligence (AI) and machine learning (ML) are increasingly being applied in health care, especially in medical imaging for diagnosing diseases.^{35,36} Images acquired via ultrasound, computed tomography, and magnetic resonance imaging, are being subjected to AI-computer-aided design (CAD) for diagnostic purposes in clinical practice. AI-CAD has shown superior or similar performance compared to conventional diagnostic approaches for various diseases including ophthalmic diseases, respiratory diseases, and cancers.³⁷ Deep learning is a form of AI with the capability to increase the accuracy as well as rapidity of the diagnosis through processing large medical image datasets, which has been deemed impossible for human experts.³⁸ Convolutional neural networks (CNNs) can classify images and detect objects in images.^{39,40,40–44} Deep learning has been used in infectious disease surveillance, including in early-warning systems for disease surveillance,⁴¹ pathogen classification,⁴² Kirby–Bauer disk diffusion assay interpretation (for bacteria),⁴³ nosocomial outbreak source identification,⁴⁴ and risk assessment.⁴⁵

Table 1. Clinical *Candida* species isolates used in this study and their GenBank accession numbers.

Species number	Isolate	GenBank accession number
1	<i>Candida albicans</i> #1	OR490529
2	<i>C. albicans</i> #2	OR490530
3	<i>C. albicans</i> #3	OR490531
4	<i>C. albicans</i> #4	OR490532
5	<i>C. albicans</i> #5	OR490533
6	<i>C. auris</i> #1	OP984814
7	<i>C. auris</i> #2	OP984815
8	<i>C. auris</i> #3	OP984816
9	<i>C. auris</i> #4	OP984817
10	<i>C. auris</i> #5	OP984818
11	<i>C. glabrata</i> #1	OR490534
12	<i>C. glabrata</i> #2	OR490535
13	<i>C. glabrata</i> #3	OR490536
14	<i>C. glabrata</i> #4	OR490537
15	<i>C. glabrata</i> #5	OR490538
16	<i>C. haemulonii</i>	OR491713

A limited number of studies have evaluated ML algorithms to identify yeast species. Data generated from the Raman spectroscopy were used to train ML models to identify *Candida* species.^{26,46} Another study used stained *C. albicans* images from cosmetic products to check for contamination.⁴⁷ Deep learning techniques have also been employed to capture *C. albicans* morphologies and explore their intermixing patterns.⁴⁸ Other studies have combined technologies such as PCR, mass spectroscopy, molecular beacons, and laser-induced breakdown spectroscopy with machine to rapidly identify yeast species.^{49–52} These tests may offer improved ‘sensitivity’ and ‘specificity’⁵³ (see ‘Definitions’ section in supplementary materials) compared to traditional methods but require further validation before widespread implementation.

In this study, we evaluate the efficiency of ML models to identify four yeast species, namely *C. albicans*, *C. auris*, *C. glabrata* (*N. glabrata*), and *C. haemulonii*. Wet-mounted images are captured by bright-field live-cell microscopy. These microscopy images are then separated into single-cell, budding-cell, and cell-group images, which are used to train and validate the ML models to identify *Candida* species of pathogenic yeast. Specifically, we train, validate, and test deep neural network models to learn hierarchical representations from imaging data by stacking multiple layers of non-linear transformations. Overall, we find that ML models- (especially the InceptionV3 model) can extract relevant features from the microscopy images to identify these yeast species.

Material and methods

Strains, media, and growth conditions

The US National Research Council’s guidelines for the Care and Use of Laboratory Animals were followed. *Candida albicans*, *C. auris*, *C. glabrata*, and *C. haemulonii* isolates were obtained from clinical samples from the Alberta Precision Laboratories (APL)—Public Health Laboratory (ProvLab). All strains and isolates (Table 1) were preserved in 25% glycerol at -80°C until further use. The strains and isolates were revived by culturing from frozen stock on Sabouraud glucose agar (SDA) plates (Millipore, Darmstadt, Germany) and incubated at 35°C for 48 h. Fresh subcultures were made on SDA

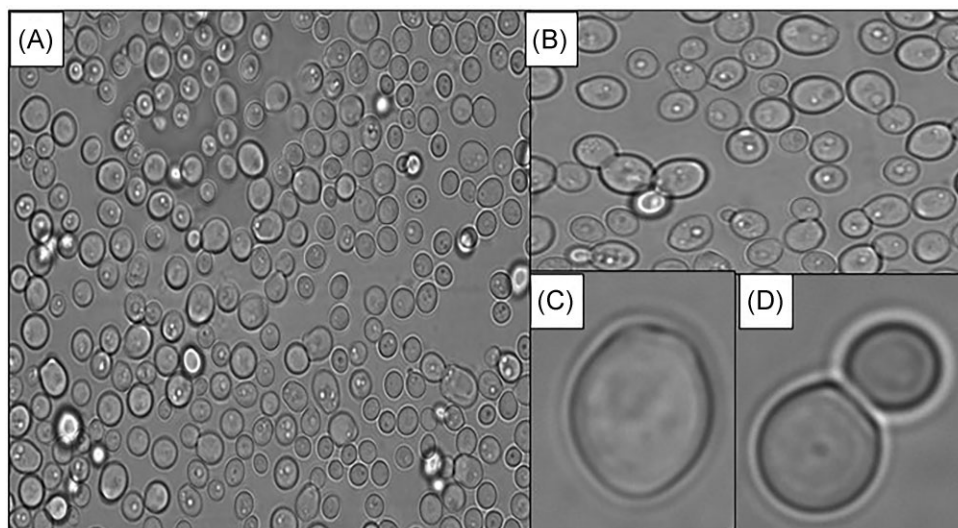


Figure 1: Images of *Candida* species captured using live-cell bright-field microscopy to train the machine learning models. (A) Raw microscopic image of *Candida* species visualized by wet-mount preparation after 24 h of growth and captured using an EVOS M7000 microscope imaging system. (B) Microscopy image showing cell groups cropped from the raw image. (C) Image showing a single cell cropped from the raw image. (D) Image showing a budding cell cropped from the raw image. All these images were used in their original aspect ratio to train the deep neural network models; some of these images appear blurry here as they have been combined with others to form a montage.

agar plates and incubated at 35°C for 24 h prior capturing high-resolution microscopy images (see ‘Data collection’ section in supplemental materials).

DNA extractions, PCR, and sequencing

All the clinical isolates were identified using MALDI-TOF MS⁵⁴ by the APL-ProvLab. Then internal transcribed spacer (ITS) region of ribosomal DNA sequencing was carried out to confirm the *Candida* species identity. The primers ITS-5 (5'-GGAAGTAAAAGTCGTAACAAGG-3') and ITS-4 (5'-TCCTCCGCTTATTGATATGC 3') were used to amplify the ITS region (Integrated DNA Technologies, Iowa, USA). Genomic DNA extraction was done using the phenol-chloroform-isoamyl alcohol method as previously described.⁵⁵ The quality and quantity of the extracted DNA were measured using a microvolume μ Drop Plate (Thermo Fisher Scientific, Mississauga, Canada). The template and the primers were mixed in concentrations of 7.5 ng/ μ l Ind 0.25 μ M, respectively, to a final volume of 10 μ l. Sanger sequencing was then performed using a 3730 Genetic Analyzer (Thermo Fisher Scientific, Mississauga, Canada) at the Molecular Biology Services Unit at the University of Alberta. The resulting sequences were subjected to nucleotide BLAST analysis,⁵⁶ which revealed 100% similarity to the standard strains. The isolates' ITS sequences were submitted to National Center for Biotechnology Information (NCBI) with the accession number OR490529-OR490538 (Table 1).

Data collection

The isolated colonies of different *Candida* species grown on SDA medium for 24 h were utilized to prepare wet mount using normal saline (0.9%) on the microscopic slide (Fisherbrand, Pittsburg, USA) overlaid with the coverslips (size: 22 mm, Fisherbrand, Pittsburg, USA). Raw images (1000 images) for each species were captured using the EVOS M7000 imaging system (Invitrogen, Thermo Fisher Scientific, Massachusetts, USA) at 100x resolution. Coarse and fine adjust-

ments were made to focus the cell on a single plane before capturing the images. These raw images were further used to crop out single cells (1000 images), budding cells (1000 images), and cell groups (1000 images) (Figure 1). A total of 4000 images per species consisting of raw images, single cells, budding cells, and cell groups saved in different directories and labelled with the name of the species (Table 2). Test sets consisting of 200 images each of raw images, single cell, budding cell, and cell groups for each species were acquired. This test set was not included either in the training or validation process. After each model was trained, a test set was used to evaluate model performance.

Preprocessing

All acquired microscopy images were scanned manually for corrupt or incomplete images, which were removed from the dataset. Images deemed problematic (e.g., images with scarce cells, out-of-focus, or blurred images, etc.) to the learning task were removed. The Python Programming Language and relevant packages (Numpy, Pandas, Matplotlib, Pillow, OpenCV, TensorFlow, Keras, Scikit-learn, and scikit-image) were used for all preprocessing, simulation, and ML. All the images were resized to be consistent with the neural network used.⁵⁷ Then the pixels in the images were scaled to the [0 1] range. This normalization helps to stabilize the training process and ensures that deep neural networks learn from similar intensity ranges across different images.⁵⁸ However, for the Inception ML model images were rescaled to the pixel values from the original range [0 255] to the range of [-1 1] or [-0.5 0.5]. Mean subtraction was done to subtract the mean red, green, and blue (RGB) value of the entire dataset from each image. Inception models often include mean subtraction as part of their preprocessing to center the data around zero, which aids in training convergence. Inception models have specific input size requirements (299 $\times$ 299 pixels), whereas EfficientNetB0 requires different image sizes (224 $\times$ 224 pixels). Images need to be resized to a fixed input size that matches the ML model's ar-

Table 2. Size of the image datasets used for training, validating, and testing each model (custom CNN, InceptionV3, VGG16, ResNet50, EfficientNetB0, and EfficientNetB7) for four different *Candida* species.

Type of image	Number of images	Images used for training (80%)	Images used for validation (20%)	Images used for testing (separate dataset)
Raw images	4000	800 ($n = 4$)	200 ($n = 4$)	200 ($n = 4$)
Single cell	4000	800 ($n = 4$)	200 ($n = 4$)	200 ($n = 4$)
Budding cell	4000	800 ($n = 4$)	200 ($n = 4$)	200 ($n = 4$)
Cell groups	4000	800 ($n = 4$)	200 ($n = 4$)	200 ($n = 4$)
Different strains of <i>Candida albicans</i> ($n = 5$) and <i>C. glabrata</i> ($n = 5$), <i>C. auris</i> ($n = 5$) (SC and BC—100 each)	3000	—	—	1000 ($n = 3$)
Total images	19 000	12 800	3200	6200

Note: A total of 16 000 images were used (1000 each of raw images, single-cell images, budding-cell images, and cell-group images for each species). Of these 16 000 images 80% (12 800 images) were used for training and 20% (3200 images) were used for validation. A separate dataset consisting of 200 images for each image type (raw images, single-cell images, budding-cell images, and cell-group images) was used for testing the models (200×4 [4 image types] $\times$ 4 [4 different *Candida* species] = 3200). Five different strains for each of *Candida albicans*, *C. glabrata*, and *C. auris* were tested for single-cell images (100 images per strain, 500 images per species) and budding-cell images (100 images per strain and 500 images per species). SC denotes single-cell and BC denotes budding-cell images.

chitecture requirements. Rescaling the images helps to reduce the impact of extreme pixel values and allows the network to converge faster.⁵⁸ Furthermore, data augmentation techniques were applied to artificially increase the diversity and quantity of training data. All the training images were subjected to data augmentation to increase the size of the dataset as well as to reduce the bias. Some of the common data augmentation techniques that we used were random rotations, translations, flips, and zooms. Data augmentation helps to reduce overfitting and improves the model's ability to generalize to unseen data.⁵⁹ Then the image datasets were split into separate subsets for training (80%) and validation (20%); a separate dataset was used for testing. The training dataset was used to train the model, and the validation dataset for hyperparameter tuning (learning rate, batch size, number of epochs, network architecture, activation functions, weight initialization, dropout rate, fine tuning, and different search strategies to explore hyperparameter space for best hyperparameters) and model selection, and the testing dataset was used to evaluate the final ML model's performance. The overall workflow to classify *Candida* species is shown in Figure 2.

Classifier architecture

Six deep CNN classifier architectures were implemented: CNN,⁶⁰ VGG-Net,⁶¹ InceptionV3,⁶² ResNet50⁶³, EfficientNetB0, and EfficientNetB7.⁶⁴ The custom CNN learning architectures consist of convolutional layers, pooling layers, and fully connected layers, which are stacked together in a sequential manner (Supplementary Figure 1). There are four convolutional layers with different filters. The first convolutional layer has 16 filters, the second has 32 filters, the third has 64 filters, and the fourth has 128 filters. All these layers use a rectified linear unit (ReLU) activation function. Following each convolutional layer, a max pooling layer is applied. After the last max pooling layer, a dropout layer with a dropout rate of 0.2 is added. Then the model flattens the 2D feature maps into a 1D vector. Two fully connected (dense) layers follow the flattened layer. The first dense layer consists of 128 units and uses the ReLU activation function. The second dense layer is the output layer with the Softmax activation function, which has four class units (once for each of the *Candida* species in our study). A total of 50 epochs were used to train the model.

The VGG16 base model was created using the VGG16 class; the weights parameter was set to 'ImageNet', which initializes the model with pre-trained weights on the ImageNet dataset.⁶⁵ Then the fully connected layers at the top of the VGG16 architecture were excluded (Supplementary Figure 2). The shape of the input images was set to $224 \times 224 \times 3$ pixels for RGB images with dimensions 224×224 pixels. The model was fine-tuned by freezing the pre-trained layers so that the learned features were retained. A new model was then created, and the base model was added as the first layer in the new model, followed by a flattening of the layer. Two fully connected (dense) layers were added after the flattened layer. The first dense layer had 64 units and the ReLU activation function was used, which introduced non-linearity to the model. The second dense layer was the output layer, which had four units corresponding to each of the *Candida* species used in this study. The SoftMax activation function was used to produce class probabilities for multi-class classification. The SoftMax function ensures that the output probabilities sum up to 1.

InceptionV3, the third version of the Inception architecture (Figure 3),⁶⁵ has fewer parameters (models with fewer parameters are often preferred for transfer learning due to faster convergence and better adaptation to new data) compared to the VGG architecture.⁶⁶ As for VGG16, images are scaled to $224 \times 224 \times 3$ pixels. Images pass through each block layer then max pooling is performed at the next layer. Images pass via these convolutional blocks and into flattened layer, which is fully linked. Four dense layers (512, 256, 128, and 4 nodes) and three dropout layers (0.5) after the first three dense layers were used. The Softmax activation function was used for the output layer.

ResNet50 stands out for its deep architecture, consisting of 50 layers (Supplementary Figure 3), which enables it to learn highly complex representations from images.⁶³ ResNet50 processes images of size $256 \times 256 \times 3$ pixels. Max pooling was used to downsample the features. After passing through the residual blocks, features were flattened to a 1D vector. Like the InceptionV3 architecture, four dense layers were utilized with the final layer consisting of four nodes representing classes. Random dropout layers were also implemented after the first three layers.

EfficientNetB0 has excellent generalization capabilities across various datasets.⁶⁴ The architecture employs a

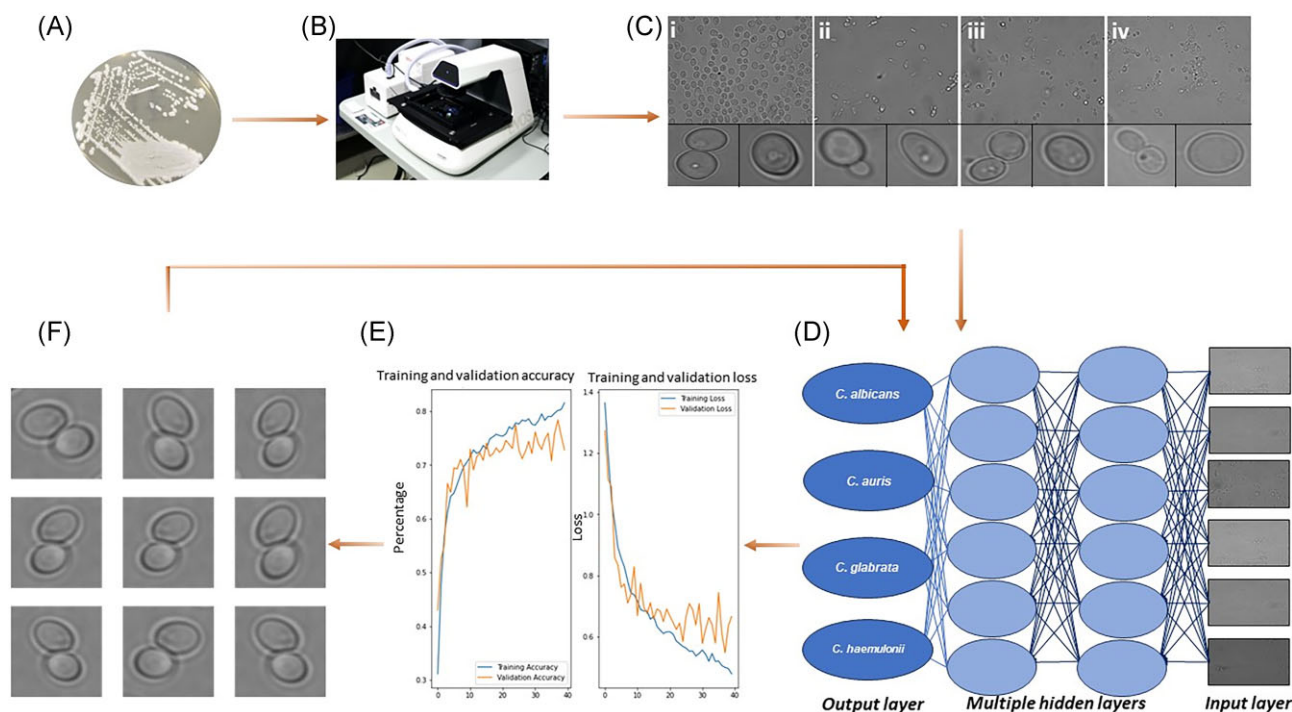


Figure 2: Schematic representation of the workflow to train machine learning models to identify four different *Candida* species. (A) *Candida albicans* on semi-solid growth medium. (B) EVOS M7000 microscope imaging system. (C) Labeled microscopy images of the different *Candida* species used in this study: (i) *C. albicans*, (ii) *C. auris*, (iii) *C. glabrata*, and (iv) *C. haemulonii* captured using an EVOS M7000 imaging system. (D) Machine learning model reading input images and passing them through the hidden layers of the convolutional neural network to extract features and output a species classification. (E) Accuracy and loss during the training and validation process. (F) Data augmentation (flipped, zoomed, rotated, and rescaled images of existing images) to reduce overfitting.

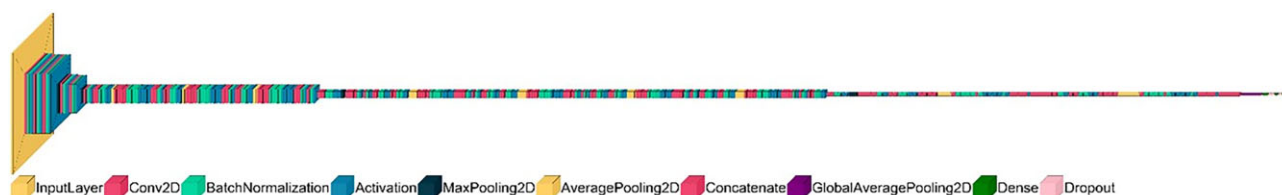


Figure 3: Schematic representation of InceptionV3 model. The color map defines the different layer types in the visualization (input, convolution, batch normalization, activation, max pooling, average pooling, concatenate, global average pooling, dense, and dropout layers).

compound scaling technique that efficiently scales up the model's dimensions while keeping computational resources in check. Whereas EfficientNetB7 represents the most advanced version of the EfficientNet CNN architecture, it has a larger model size and complexity compared to EfficientNetB0. Despite its increased computational cost, EfficientNetB7 maintains the inherent efficiency that is characteristic of the EfficientNet architecture.

Results

We evaluated the effectiveness of the six CNN models to identify four different *Candida* species. First, we evaluated the custom CNN model with and without data augmentation, whereas all other CNN models were evaluated with data augmentation. Data augmentation artificially increases the number of images, improves model generalization, reduces overfitting and variance to transformations, and overall improves performance on real-world data. The training accuracy without data augmentation for the custom CNN model

reached up to 99.6%, whereas the validation accuracy was 66.9%. Post-data augmentation, the training and validation accuracy were respectively 85.4% and 83.9%, respectively (Supplementary Figure 4). However, the precision and recall for this model were 25.0% and 38.2%, respectively. Overall, the performance of the custom CNN model on the unseen test dataset was not satisfactory (Supplementary Table 1). The raw images contain blank spaces, which this model received as 'garbage data' (i.e., data without any value towards predicting the output). Therefore, we cropped out images of single, budding, and cell groups from the raw images and then trained the model (see 'Data Collection' section in supplemental materials). This model trained on the augmented data was able to identify *Candida albicans* on raw cell images and budding cells respectively at 75.0% and 82.0%, though at the cost of poorly identifying *C. albicans* from single-cell images (12.5%). Overall, this model performed poorly in identifying *C. haemulonii* in all the testing datasets (Supplementary Table 2).

Next, we trained the ResNet50 model with data augmentation. With data augmentation, the training and validation

Table 3. Results of the InceptionV3 model and for each class trained on single and budding cell images.

Metrics	InceptionV3	<i>Candida albicans</i>	<i>C. auris</i>	<i>C. glabrata</i>	<i>C. haemulonii</i>
Accuracy (%)	74.6	74.6	74.6	74.6	74.6
Sensitivity (%)	77.1	88.5	63.0	55.0	91.2
Specificity (%)	92.4	86.2	96.2	96.1	87.5
Precision (%)	77.9	68.2	84.8	82.8	70.8
Negative predictive value (%)	92.4	95.7	88.6	86.7	96.7
F ₁ score (%)	77.14	77.0	72.3	66.6	79.7
AUC (%)	91.9	94.3	87.0	90.1	96.1
Time to identify (ms)	-	0.02	0.02	0.02	0.02

Note: See 'Definitions' section for quantitative definitions of each metric.

accuracy respectively reached 98.7% and 73.0% at the end of epoch 20 (Supplementary Figure 5). However, the precision and recall for each class was <30.0% (Supplementary Table 4). The test set containing the raw images was used to test this model's predictive ability. Trained ResNet50 was able to predict all the *C. albicans* (100.0%) raw images. However, it could only predict *C. haemulonii*, *C. glabrata*, and *C. auris* in 18.0%, 6.9%, and 0.5% of the cases, respectively. We then used the test set containing the single-cell images. The model correctly predicted *C. albicans*, *C. auris*, *C. glabrata*, and *C. haemulonii* in 88.5%, 71.0%, 62.5%, and 40.0% instances, respectively. The model was able to predict a test set containing *C. albicans* budding cells at 96.0%, followed by *C. haemulonii* (61%), *C. glabrata* (28.4%), and *C. auris* (28.4%).

Then we trained the InceptionV3 model with data augmentation. The training and validation accuracy reached 92.4% and 78.7%, respectively. The precision and recall for this model trained and validated on all the image sets are provided in Supplementary Table 5. When we evaluated the trained InceptionV3 model with a test set containing raw images this model was able to correctly classify the majority of *C. albicans* images (95.5%). However, this model could not identify *C. auris* and *C. haemulonii*, as the images were misclassified as *C. albicans*. A low success rate of 29.2% was also observed in correctly classifying the *C. glabrata* images. When the test set containing only budding cells was evaluated this model was able to correctly identify *C. albicans*, *C. auris*, *C. glabrata*, and *C. haemulonii* budding cells in 87.7%, 20.0%, 77.0%, and 82.3% of cases, respectively. Whereas, in the case of test set images of single cell belonging to *C. albicans*, *C. auris*, *C. glabrata*, and *C. haemulonii* were correctly identified in 87.0%, 59.0%, 85.0%, and 36.5% of cases, respectively. Then we trained this model using images of single cell and budding cell belonging to different classes and found that the training and validation accuracies reached 76.3% and 74.6%, respectively, at the end of epoch 50. We then evaluated other metrics (negative predictive value, F1 score, AUC score, and time to identify) for this model (Table 3). The receiver operator characteristic (ROC) curve is a plot that depicts the trade-off between the sensitivity and specificity across a series of cutoff points when the diagnostic test is continuous or on an ordinal scale.⁶⁷ The ROC curve for the different *Candida* species is presented in Figure 4. When we tested the model's performance on the test set containing single cells and budding cells of different *Candida* species this model could respectively identify budding cells of *C. albicans*, *C. auris*, *C. glabrata*, and *C. haemulonii* in 97.0%, 74.0%, 68%, and 66% of the cases.

Similarly, for single-cell test set model could respectively identify *C. albicans*, *C. auris*, *C. glabrata*, and *C. haemulonii* in 97.0%, 73.0%, 69%, and 73% of the cases.

We then evaluated the VGG16 model to determine if it could outperform the InceptionV3 model. When the VGG16 model was trained on all the image datasets, the overall training accuracy was 73.0% (Supplementary Table 6). We found a similar accuracy (73.7%) when the VGG16 model trained on single-cell and budding-cell image dataset alone (Supplementary Table 7). EfficientNetB0 and EfficientNetB7 models were also trained for raw images as well as single and budding cell images. However, the precision and recall for both models were below 25.0%.

Finally, we tested the ability of the InceptionV3 model to identify five different strains of the same species for *C. albicans*, *C. glabrata*, and *C. auris*. Of the total 500 images of single cells and budding cells each belonging to each strain (100 images of single cells and budding cells for each strain) the trained InceptionV3 model was able to identify accurately single-cell images of *C. albicans*, *C. glabrata*, and *C. auris* in 93.4%, 69.2%, and 73.5% of the cases, respectively. Whereas for the budding cell images, the model was able to identify *C. albicans*, *C. glabrata*, and *C. auris* in 92.0%, 75.4%, and 68.3% of the cases.

Discussion

Image-based studies incorporating ML algorithms have successfully been implemented to identify disease states, quantify biomarkers, detect mitosis, recognize lymph node metastasis, tissue segmentation, prognostication, and to predict molecular expression and treatment responses.⁶⁸ Our proof-of-concept study demonstrates that overall microscopy images obtained using wet-mounted slides can be used to identify different *Candida* species using ML models. However, our results indicate that black spaces between cells in the raw (unprocessed) images did not add to useful information with respect to cell features. In contrast, single-cell and budding-cell images with minimal blank spaces proved ideal. Different *Candida* species have different cell size ranges and budding pattern,⁶⁹ which may have been critical for our ML models to differentiate between the species. Our model was able to distinguish *Candida albicans* more efficiently from other tested *Candida* species. One hypothesis for the misclassification of *C. glabrata* and *C. haemulonii* as *C. albicans* is due to the pleomorphic nature of *C. albicans*. The varying size and shape of *C. albicans* are learned by the ML models and while classifying other *Candida* species some of these features might resemble that

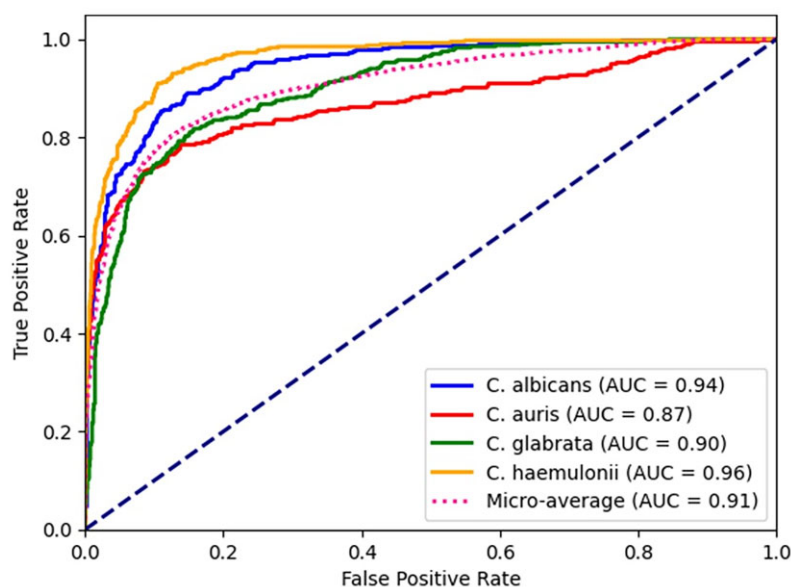


Figure 4: AUC–ROC curve of InceptionV3 model and AUC value with individual class scores. Dashed blue line (no-discrimination line) represents the performance of a random guessing classifier.

of *C. albicans*, which may lead to misclassification.⁶⁹ Further studies using stained cell images to train the ML models could help overcome this issue to attain a score of >99% akin to an experienced microbiologist.⁷⁰

Performance of different ML algorithms depends on the dataset used during the learning process. Similarly, our study showed that the microscopy image dataset we generated was effectively learned by the InceptionV3, which was able to classify the species of *Candida* most accurately. The high performance of InceptionV3 compared to other models has been reported previously. For instance, InceptionV3 showed high accuracy and sensitivity to diagnose poultry diseases.⁷¹ Another study on ankle fracture detection using deep learning algorithms found that InceptionV3 model had high specificity and sensitivity.⁷² Similarly, InceptionV3 was able to identify skin cancer-melanoma with high accuracy.⁷³ In contrast, Rahman et al., reported that the InceptionV3 model could reach an accuracy of 61.4% with augmented data while classifying 89 different fungal genera.⁷⁴ One reason for the low accuracy of InceptionV3 in their study may be due to the relatively small microscopy imaging dataset (1079 images of 89 genera). However, in our study custom CNN, VGG16, ResNet50, EfficientNetB0, and EfficientNetB7 were not as efficient as InceptionV3 in classifying *Candida* species. Similar instances of poor performance of these deep neural networks have also been found previously.^{75–77} Some of these models performed well on the other datasets such as ImageNet, CIFAR (Canadian Institute for Advanced Research), and MNIST (Modified National Institute of Standards and Technology) databases but failed to learn about the microscopy images in this study. In our study, we showed that the trained model was able to identify the images within 0.02 milliseconds (Table 3). In contrast, diagnostic labs inoculate clinical samples onto agar plate or automated blood culture system (blood samples) and the resulting growth after incubating for 18–24 h is utilized for the identification.⁷⁸ Identifying the etiologic agent may further take around 24–48 h.⁷⁸ Whereas the sample preparation method used in this study is comparatively easy with limited

required resources after we obtain the yeast colonies on the agar plate.

One common challenge with ML models is their ‘black box’ nature, which makes it difficult to understand how they arrive at their predictions.⁷⁹ This lack of interpretability hinders their adoption in clinical practice where transparency and trust are crucial. To address this challenge, researchers in the field of explainable AI have explored different approaches to make ML models more interpretable.^{80–84} For instance, by generating heatmaps or saliency maps, identifying informative features through feature reduction or selection, or providing immersive visualizations through VR technology, researchers can enhance transparency and trust in ML predictions. We speculate that the subtle uniqueness of each species in their shape, size, and the budding pattern along with the budding base might have contributed as the important features for the ML process. Similarly, further exploration of the model’s ability to predict different *Candida* species will provide more insight into the features of the images required to identify them accurately. Based on this proof of concept study, we can further elaborate on the use of these ML models in many areas such as (1) extrapolating the application of the ML approach to detect other yeast species, which in turn can help its diagnostic utility, (2) employing different complex models (e.g., DenseNet, Xception, MobileNetV2, ResNeXt, and SENet⁸⁵) to improve performance metrics, (3) location-specific training of the yeast images is required for its use, and (4) differentiating antifungal resistant isolates from susceptible isolates. One of the limitations of our study is that we did not consider all the yeast species causing infection (e.g., *Pichia kudriavzevii* [*C. krusei*], *C. parapsilosis*, and *C. tropicalis*); we considered strains from four *Candida* species: *C. albicans*, *C. auris*, *C. glabrata* (*Nakaseomyces glabrata*), and *C. haemulonii*. Thus, it is possible that our ML approach may face challenges in identifying morphologically similar species, though we do not anticipate these challenges for morphologically distinct fungi. However, even closely related (phylogenetically and morphologically) fungal species differ in several physio-

logical and biochemical properties,⁸⁶ as well as their microbiological species-specific staining profiles.⁸⁷ Utilizing ML methods to leverage these distinctions may aid in detecting morphologically similar species, which is a promising direction for future research. We also did not test microscopy images obtained from co-culturing different *Candida* species. In clinical settings, multi-species infections are encountered and are challenging to treat.⁸⁸ Distinguishing yeast species from other cell types and components of clinical samples such as blood and urine would further elaborate the utility of our method for clinical applications. Nevertheless, we anticipate that our ML-based approach, which does not require sophisticated instruments or extensive expertise to implement, will help to reduce the turnaround times for the diagnosis of yeast infections.

Our study demonstrates that the identification time for *Candida* species can be drastically reduced and requires less expertise unlike for sophisticated techniques such as MALDI-TOF MS or nucleic acid-based amplification tests. Our study also highlights the ease of use in exploiting the ML approach for rapid identification of *Candida* species using microscopy images. To optimize the detection capability of our ML models, we separated out images of single cells, groups of cells, and budding cells from the raw microscopy images. However, in a clinical microbiology laboratory separation of individual cells and budding cells from the captured microscopic images adds a step to the workflow before image classification. This can be overcome by automating the image separation process with unsupervised ML,⁷⁹ where the individual cells, groups of cells, and budding cells can be cropped based on the contours, borders, and intensity from the raw images.⁸⁹ Considering the rapid emergence of antifungal-resistant species,³³ it will be indispensable to evaluate the ability of these ML models to determine susceptibility profiles in future work. For instance, data from the whole genome sequence of *C. auris* has been used to train ML models to predict the mutations responsible for antifungal drug resistance, rank different resistance mutations, and discover new mutations related to drug resistance.⁹⁰ Overall, we anticipate that ML approaches will be able to detect antifungal-resistant phenotypes, elucidate underlying molecular resistance mechanisms, and improve the treatment of patients with infectious fungal diseases.

Declaration of interest

The authors report no conflicts of interest. The authors alone are responsible for the content and the writing of the paper.

Supplementary material

Supplementary material is available at *Medical Mycology* online.

Acknowledgements

We thank Dr. Tanis Dingle and the Alberta Precision Laboratories-Public Health Laboratory for providing the *Candida auris* isolates. We acknowledge the Samira Rasouli Koohi and the Molecular Biology Services Unit at the University of Alberta for assistance with genetic sequencing. We acknowledge Prof. Randy Goebel, Prof. Jay Newby, and Jiyai Dai for guidance on choosing, training, and validating the machine learning models. We thank Joshua Guthrie for technical support and proofreading the manuscript.

Author contributions

Shamanth A. Shankarnarayan (Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing), and Daniel A. Charlebois (Conceptualization, Funding acquisition, Methodology, Supervision, Writing – original draft, Writing – review & editing)

Funding

This work was supported by the Canadian Foundation for Innovation's John R. Evans Leaders Fund (CFI-40558), the Government of Alberta's Research Capacity Program (RCP-21-008-SEG), and seed grants from AI4Society and the Alberta Machine Intelligence Institute, and an Audrey and Randy Lomnes Early Career Endowment Award to D.C.

Data availability

The data that support the findings of this study are available upon reasonable request from the corresponding author.

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